Analysis

April 14, 2026

0.0.1 0. Library Imports

[1]:
```
import re
import random
import pandas as pd
import numpy as np
import csv
from pathlib import Path

import math
from collections import defaultdict
from scipy.sparse import lil_matrix

from sklearn.svm import SVC
from sklearn.linear_model import LogisticRegression
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.preprocessing import StandardScaler
from scipy.sparse import hstack

import torch
from torch.optim import Adam
from torch.utils.data import Dataset, DataLoader

from transformers import BertTokenizer, BertModel, BertTokenizerFast,␣
↪BertForSequenceClassification

import gensim.downloader as api
from gensim.models import Word2Vec, FastText

import matplotlib.pyplot as plt
import seaborn as sns
```

0.0.2 1. Set Device type

[2]:
```
if torch.cuda.is_available():
    device = torch.device("cuda")
elif torch.backends.mps.is_available():
    device = torch.device("mps")
else:
    device = torch.device("cpu")

print(f"Using device: {device}")
```

```
Using device: mps
```

0.0.3 2. Load Dataset

[3]:
```
df = pd.read_csv('dataset/support-ticket-classification.csv')

y = np.array(df['label'].values)
unique_labels = sorted(df['label'].unique().tolist())

print(f"Dataset loaded: {df.shape}")
print(f"Classes: {unique_labels}")
print(f"Class distribution:\n{df['label'].value_counts()}\n")
```

```
Dataset loaded: (2000, 4)
Classes: ['account_access', 'billing', 'bug_report', 'refund_request',
'shipping_delivery']
Class distribution:
label
account_access       400
billing              400
refund_request       400
bug_report           400
shipping_delivery    400
Name: count, dtype: int64
```

0.0.4 3. Preprocessing Steps

[4]:
```
def load_stopwords(file_path):
    stopwords = set()
    with open(file_path, 'r', encoding='utf-8') as f:
        for line in f:
            word = line.strip().lower()
            if word and not word.startswith('#'):
                stopwords.add(word)
    return stopwords
```

```python
stopwords = load_stopwords('dataset/stopwords.txt')

for w in ['the', 'is', 'account', 'billing', 'login', 'and', 'password']:
    print(f"{w} -> stopword: {w in stopwords}")
```

```
the -> stopword: True
is -> stopword: True
account -> stopword: False
billing -> stopword: False
login -> stopword: False
and -> stopword: True
password -> stopword: False
```

[5]:
```python
def load_exc_files(noun_file, verb_file):
    exc_map = {}
    for file_path in [noun_file, verb_file]:
        if not Path(file_path).exists():
            continue
        with open(file_path, 'r', encoding='utf-8') as f:
            for line in f:
                parts = line.strip().split()
                if len(parts) >= 2:
                    exc_map[parts[0]] = parts[1]
    return exc_map

exc_map = load_exc_files('dataset/wordnet_noun.exc', 'dataset/wordnet_verb.exc')

for word in ['men', 'mice', 'geese', 'ran', 'is']:
    if word in exc_map:
        print(f"{word} -> {exc_map[word]}")
```

```
men -> man
mice -> mouse
geese -> goose
ran -> run
is -> be
```

[6]:
```python
def load_lemma_dict(file_path):
    lemma_map = {}
    with open(file_path, 'r', encoding='utf-8') as f:
        for line in f:
            line = line.strip()
            if not line or line.startswith(';') or ' -> ' not in line:
                continue
            lemma_part, _, forms_part = line.partition(' -> ')
            lemma = lemma_part.split('/')[0].strip()
            for form in forms_part.split(','):
                form = form.strip()
```

```
                if form and re.match(r'^[a-z0-9]+$', form) and form not in␣
↪lemma_map:
                    lemma_map[form] = lemma
    return lemma_map

lemma_map = load_lemma_dict('dataset/lemma.en.txt')
print(f"lemma_map loaded: {len(lemma_map)} entries\n")

for word in ['running', 'categories', 'responses', 'presses']:
    print(f"{word} -> {lemma_map.get(word, '(not found)')}")
```

```
lemma_map loaded: 90502 entries

running -> run
categories -> category
responses -> response
presses -> press
```

[7]:
```
CONTRACTIONS = {
    "can't": "cannot", "won't": "will not", "don't": "do not",
    "doesn't": "does not", "didn't": "did not", "isn't": "is not",
    "aren't": "are not", "wasn't": "was not", "weren't": "were not",
    "i'm": "i am", "you're": "you are", "we're": "we are",
    "they're": "they are", "it's": "it is", "that's": "that is",
    "there's": "there is", "i've": "i have", "we've": "we have",
    "they've": "they have", "shouldn't": "should not", "couldn't": "could not",
    "wouldn't": "would not", "hasn't": "has not", "haven't": "have not",
    "hadn't": "had not",
}

for k, v in list(CONTRACTIONS.items())[:3]:
    print(f"{k} -> {v}")
```

```
can't -> cannot
won't -> will not
don't -> do not
```

[8]:
```
DOMAIN_PHRASE_MAP = {
    "log in": "login", "sign in": "login",
    "wireless": "wifi", "wifi": "wifi",
    "remote access": "vpn", "vpn": "vpn",
    "mfa": "mfa", "2fa": "mfa",
}

DOMAIN_TOKEN_MAP = {
    "pls": "please", "plz": "please", "asap": "urgent",
    "u": "you", "ur": "your", "pwd": "password",
    "acct": "account", "acc": "account", "auth": "authentication",
```

```
    "db": "database", "srv": "server", "msg": "message",
    "req": "request", "otp": "onetimepassword", "2fa": "mfa",
}

for k, v in list(DOMAIN_PHRASE_MAP.items())[:3]:
    print(f"phrase  {k} -> {v}")
for k, v in list(DOMAIN_TOKEN_MAP.items())[:3]:
    print(f"token   {k} -> {v}")
```

```
phrase  log in -> login
phrase  sign in -> login
phrase  wireless -> wifi
token   pls -> please
token   plz -> please
token   asap -> urgent
```

[9]:
```
NEGATION_TOKENS = {"no", "not", "never", "cannot", "failed", "unable",␣
 ↪"without"}
DOMAIN_STOPWORDS = {"hi", "hello", "dear", "team", "please", "kindly",␣
 ↪"thanks", "thank", "regards", "best", "sir", "madam"}
ALLOWED_TOKEN_PUNCT = set(".-/@#+_:\\")

for w in ['cannot', 'not', 'hello', 'billing', 'please', 'account']:
    print(f"{w}  negation={w in NEGATION_TOKENS}  stopword={w in␣
 ↪DOMAIN_STOPWORDS}")
```

```
cannot  negation=True  stopword=False
not  negation=True  stopword=False
hello  negation=False  stopword=True
billing  negation=False  stopword=False
please  negation=False  stopword=True
account  negation=False  stopword=False
```

[10]:
```
REPLY_CHAIN_PATTERN = re.compile(r"^\s*on\s+.+\s+wrote:\s*$", flags=re.
 ↪IGNORECASE)
HEADER_PATTERN = re.compile(r"^\s*(from|sent|to|cc|bcc|date)\s*:", flags=re.
 ↪IGNORECASE)
SIGNATURE_PATTERN = re.compile(
    r"^\s*(thanks|thank you|regards|best regards|kind␣
 ↪regards|sincerely|cheers)[\s,!.-]*$",
    flags=re.IGNORECASE,
)
DISCLAIMER_PATTERN = re.compile(
    r"(this\s+(e-?mail|message).{0,120}(confidential|intended␣
 ↪only)|please\s+do\s+not\s+reply|"
    r"automatically\s+generated|auto-?generated)",
    flags=re.IGNORECASE,
)
```

```
samples = [
    ("On Mon Apr 12 user@x.com wrote:", REPLY_CHAIN_PATTERN, "reply_chain"),
    ("From: user@example.com",          HEADER_PATTERN,      "header"),
    ("Thanks,",                          SIGNATURE_PATTERN,   "signature"),
    ("This email is auto-generated",     DISCLAIMER_PATTERN,  "disclaimer"),
]
for text, pattern, label in samples:
    print(f"{label} {' ' if pattern.search(text) else ' '}  {text}")
```

```
reply_chain    On Mon Apr 12 user@x.com wrote:
header    From: user@example.com
signature    Thanks,
disclaimer    This email is auto-generated
```

[11]:
```
ENTITY_TAG_PATTERNS = [
    ("IP_ADDR",    re.compile(r"\b(?:\d{1,3}\.){3}\d{1,3}\b")),
    ("TICKET_ID",  re.compile(r"\b(?:inc|ticket|case|ref|req)[-_:
 ↪#\s]*\d{3,}\b", flags=re.IGNORECASE)),
    ("ORDER_ID",   re.compile(r"\b(?:order|ord)[-_:#\s]*\d{3,}\b", flags=re.
 ↪IGNORECASE)),
    ("VERSION",    re.compile(r"\bv\d+(?:\.\d+){1,3}\b", flags=re.IGNORECASE)),
    ("ERROR_CODE", re.compile(r"\b(?:0x[a-fA-F0-9]{4,}|(?:
 ↪error|err|errno|exception)[-_:#\s]*[a-z0-9_-]{2,})\b", flags=re.IGNORECASE)),
    ("LONG_NUM",   re.compile(r"\b\d{6,}\b")),
]

TOKEN_PATTERN = re.compile(
    r"(?:https?://\S+|www\.\S+)|"
    r"(?:[A-Za-z]:\\[^\s]+)|"
    r"(?:[\w.+-]+@[\w.-]+\.\w+)|"
    r"(?:\b\w+(?:[._:/\\#@+-]\w+)+\b)|"
    r"(?:\b\w+\b)"
)

sample = "Error 0x1234ABCD on ticket INC-5432 from 192.168.1.1, order 98765, v1.
 ↪2.3"
for tag, pattern in ENTITY_TAG_PATTERNS:
    m = pattern.search(sample)
    if m:
        print(f"{tag} matched: {m.group(0)}")
```

```
IP_ADDR matched: 192.168.1.1
TICKET_ID matched: INC-5432
ORDER_ID matched: order 98765
VERSION matched: v1.2.3
ERROR_CODE matched: Error 0x1234ABCD
```

[12]:
```python
def normalize_whitespace(text, keep_newlines=False):
    if keep_newlines:
        text = re.sub(r"[^\S\n]+", " ", text)
        text = re.sub(r"\n{3,}", "\n\n", text)
        return text.strip()
    return re.sub(r"\s+", " ", text).strip()

def clean_text(text):
    if pd.isna(text):
        return ""
    text = str(text)
    text = text.replace("\r\n", "\n").replace("\r", "\n")
    text = text.replace("\u200b", " ")
    text = re.sub(r"[\t\f\v]+", " ", text)
    return normalize_whitespace(text, keep_newlines=True)

def remove_boilerplate(text):
    lines = [line.strip() for line in text.split("\n")]
    kept_lines = []
    for idx, line in enumerate(lines):
        if not line:
            continue
        if REPLY_CHAIN_PATTERN.match(line):
            break
        if HEADER_PATTERN.match(line):
            continue
        if DISCLAIMER_PATTERN.search(line):
            continue
        if SIGNATURE_PATTERN.match(line) and idx >= max(2, len(lines) // 3):
            break
        kept_lines.append(line)
    return normalize_whitespace("\n".join(kept_lines))

sample_email = "From: user@example.com\nDate: Mon Apr 12\n\nI cannot login to␣
↪my account.\n\nThanks\n\nOn Mon Apr 12 support@company.com wrote:\n>␣
↪Previous message here."
print("raw   :", repr(sample_email[:80]))
print("clean :", repr(clean_text(sample_email)[:80]))
print("boiler:", repr(remove_boilerplate(clean_text(sample_email))[:80]))
```

```
raw   : 'From: user@example.com\nDate: Mon Apr 12\n\nI cannot login to my
account.\n\nThanks\n\n'
clean : 'From: user@example.com\nDate: Mon Apr 12\n\nI cannot login to my
account.\n\nThanks\n\n'
boiler: 'I cannot login to my account.'
```

[13]:
```
def expand_contractions(text):
    for contraction, expanded in CONTRACTIONS.items():
        text = re.sub(rf"\b{re.escape(contraction)}\b", expanded, text,␣
↪flags=re.IGNORECASE)
    return text

def apply_domain_mapping(text, phrase_map=None):
    phrase_map = phrase_map or DOMAIN_PHRASE_MAP
    for source, target in phrase_map.items():
        text = re.sub(rf"\b{re.escape(source)}\b", target, text, flags=re.
↪IGNORECASE)
    return text

samples = [
    "I can't log in with 2fa",
    "Won't connect over wireless",
    "Please sign in using remote access",
]
for s in samples:
    print(f"{s} -> {apply_domain_mapping(expand_contractions(s))}")
```

```
I can't log in with 2fa -> I cannot login with mfa
Won't connect over wireless -> will not connect over wifi
Please sign in using remote access -> Please login using vpn
```

[14]:
```
def normalize_entities(text, add_tags=True):
    if not add_tags:
        return text
    for tag, pattern in ENTITY_TAG_PATTERNS:
        text = pattern.sub(lambda m: f" {m.group(0)} {tag} ", text)
    return normalize_whitespace(text)

def tokenize(text):
    return TOKEN_PATTERN.findall(text)

def normalize_token(token, lowercase=True):
    token = token.strip("\"'`()[]{}<> ").strip(".,;!?")
    if not token:
        return ""
    if token.isalpha():
        token = re.sub(r"(.)\1{2,}", r"\1\1", token)
    mapped = DOMAIN_TOKEN_MAP.get(token.lower())
    if mapped is not None:
        return mapped
    return token.lower() if lowercase else token

def is_identifier_like(token):
```

```
    if not token:
        return False
    if token.isupper() and len(token) > 1:
        return True
    if any(ch.isdigit() for ch in token):
        return True
    if any(ch in ALLOWED_TOKEN_PUNCT for ch in token):
        return True
    return False

sample = "Error 0x1234ABCD on ticket INC-5432 from 192.168.1.1, pls fix asap"
print("entities :", normalize_entities(sample))
print("tokens   :", tokenize(sample)[:8])
print("normed   :", [normalize_token(t) for t in tokenize(sample)[:6]])
```

```
entities : Error 0x1234ABCD ERROR_CODE on ticket INC-5432 TICKET_ID from
192.168.1.1 IP_ADDR , pls fix asap
tokens   : ['Error', '0x1234ABCD', 'on', 'ticket', 'INC-5432', 'from',
'192.168.1.1', 'pls']
normed   : ['error', '0x1234abcd', 'on', 'ticket', 'inc-5432', 'from']
```

[15]:
```
def should_lemmatize(token):
    if token in NEGATION_TOKENS:
        return False
    if not token.isalpha() or len(token) < 4:
        return False
    if is_identifier_like(token):
        return False
    if token.endswith(("ing", "ed")):
        return False
    return True

def postprocess_tokens(tokens, lemma_map=None, exc_map=None,␣
 ↪remove_stopwords=True, stopwords=None):
    if stopwords is None:
        stopwords = set()
    processed = []
    for token in tokens:
        token = normalize_token(token, lowercase=True)
        if len(token) <= 1:
            continue
        if remove_stopwords and token in stopwords and token not in␣
 ↪NEGATION_TOKENS:
            continue
        if should_lemmatize(token):
            lemma = exc_map.get(token) if exc_map else None
            if lemma is None and lemma_map:
```

```
                lemma = lemma_map.get(token)
            if lemma:
                token = lemma
        if remove_stopwords and token in stopwords and token not in␣
 ↪NEGATION_TOKENS:
            continue
        processed.append(token)
    return processed

tokens = ['accounts', 'the', 'categories', 'pls', 'cannot', 'responses']
result = postprocess_tokens(tokens, lemma_map=lemma_map, exc_map=exc_map,␣
 ↪remove_stopwords=True, stopwords=DOMAIN_STOPWORDS)
print(f"before : {tokens}")
print(f"after  : {result}")
```

```
before : ['accounts', 'the', 'categories', 'pls', 'cannot', 'responses']
after  : ['account', 'the', 'category', 'cannot', 'response']
```

[16]:
```
def remove_boilerplate(text):
    lines = text.split('\n')
    content_lines = []
    for line in lines:
        if REPLY_CHAIN_PATTERN.match(line):
            continue
        if HEADER_PATTERN.match(line):
            continue
        if SIGNATURE_PATTERN.search(line):
            continue
        if DISCLAIMER_PATTERN.search(line):
            continue
        content_lines.append(line)
    return '\n'.join(content_lines).strip()

def apply_domain_mapping(text):
    for phrase, replacement in DOMAIN_PHRASE_MAP.items():
        text = re.sub(rf'\b{re.escape(phrase)}\b', replacement, text, flags=re.
 ↪IGNORECASE)
    for token, replacement in DOMAIN_TOKEN_MAP.items():
        text = re.sub(rf'\b{re.escape(token)}\b', replacement, text, flags=re.
 ↪IGNORECASE)
    return text

def tokenize(text):
    return re.findall(r'\b\w+\b', text.lower())

def postprocess_tokens(tokens, lemma_map):
    processed = []
```

```
    for token in tokens:
        if token not in stopwords_set and should_lemmatize(token):
            lemmatized = morphy(token, lemma_map, exc_map)
            if lemmatized:
                processed.append(lemmatized)
        elif token not in stopwords_set:
            processed.append(token)
    return processed

def morphy(word, lemma_map, exc_map):
    if word in lemma_map:
        return lemma_map[word]
    if word in exc_map:
        return exc_map[word]
    return word

DISCLAIMER_PATTERN = re.compile(r'(auto-generated|confidential|disclaimer)',␣
 ↪flags=re.IGNORECASE)

stopwords_set = load_stopwords('dataset/stopwords.txt')
exc_map = load_exc_files('dataset/wordnet_noun.exc', 'dataset/wordnet_verb.exc')
lemma_map = load_lemma_dict('dataset/lemma.en.txt')
```

[17]:
```
def preprocess_pipeline(text, for_embedding=False):
    text = clean_text(text)
    text = remove_boilerplate(text)
    text = expand_contractions(text)
    text = apply_domain_mapping(text)
    text = normalize_entities(text, add_tags=False)
    text = normalize_whitespace(text)
    if not for_embedding:
        text = text.lower()
    tokens = tokenize(text)
    if for_embedding:
        return ' '.join(tokens)
    processed_tokens = postprocess_tokens(tokens, lemma_map)
    return ' '.join(processed_tokens)

print("Creating preprocessing columns...")
df['raw_text'] = df['subject'].fillna('') + ' ' + df['text'].fillna('')
df['tfidf_text'] = df['raw_text'].apply(lambda x: preprocess_pipeline(x))
df['embedding_text'] = df['raw_text'].apply(lambda x: preprocess_pipeline(x,␣
 ↪for_embedding=True))
df['fasttext_text'] = df['raw_text'].apply(lambda x: preprocess_pipeline(x))
df['bert_text'] = df['raw_text'].apply(lambda x: preprocess_pipeline(x))

print(f"Columns created: {list(df.columns)}")
```

```
print(f"Example raw_text: {df['raw_text'].iloc[0][:100]}")
print(f"Example tfidf_text: {df['tfidf_text'].iloc[0][:100]}")

x_train, x_test, y_train, y_test = train_test_split(
    np.arange(len(df)), y, test_size=0.3, stratify=y
)

print(f"Train/test split: {len(x_train)} train, {len(x_test)} test")
print(f"y_train shape: {y_train.shape}, y_test shape: {y_test.shape}")
```

```
Creating preprocessing columns…
Columns created: ['id', 'label', 'subject', 'text', 'raw_text', 'tfidf_text',
'embedding_text', 'fasttext_text', 'bert_text']
Example raw_text: Guest access permisons wrong Guest access isn't respecting the
permission levels I set.
Example tfidf_text: guest access permisons wrong guest access respecting
permission level set
Train/test split: 1400 train, 600 test
y_train shape: (1400,), y_test shape: (600,)
```

[18]:
```
def analyze_misclassified(method_name, y_pred, X_test_indices, y_test_true,␣
↪max_per_class=3):
    import numpy as np

    y_test_array = y_test_true if isinstance(y_test_true, np.ndarray) else␣
↪y_test_true.values
    misclassified_idx = np.where(y_pred != y_test_array)[0]

    print(f"\n{method_name}")
    print("-" * 100)

    if len(misclassified_idx) == 0:
        print("No misclassified examples!")
        return

    for class_label in sorted(unique_labels):
        class_errors = [idx for idx in misclassified_idx if y_test_array[idx]␣
↪== class_label]
        if not class_errors:
            continue

        selected = class_errors[:max_per_class]
        print(f"\nClass '{class_label}': {len(class_errors)} errors (showing␣
↪{len(selected)})")

        for i, idx in enumerate(selected, 1):
            text = df.iloc[X_test_indices[idx]]['text'][:150].replace("\n", " ")
```

```
            print(f"{i}. Actual={y_test_array[idx]} | Pred={y_pred[idx]} | Text:
↪ {text}...")
```

0.0.5 4. TF-IDF

[19]:
```
class TFIDFExtractor:
    def __init__(self, word_max=35000, min_df=2, max_df=0.95, ngram_range=(1,␣
↪2)):
        self.word_max = word_max
        self.min_df = min_df
        self.max_df = max_df
        self.ngram_range = ngram_range
        self.vocab = {}
        self.idf = {}
        self.n_docs = 0

    def _get_ngrams(self, tokens, n):
        return [' '.join(tokens[i:i+n]) for i in range(len(tokens) - n + 1)]

    def _tokenize(self, text):
        text = text.lower()
        tokens = re.findall(r'\b\w+\b', text)
        return tokens

    def fit(self, texts):
        doc_freq = defaultdict(int)
        n_docs = len(texts)
        self.n_docs = n_docs

        for text in texts:
            tokens = self._tokenize(text)
            seen_terms = set()

            for n in range(self.ngram_range[0], self.ngram_range[1] + 1):
                ngrams = self._get_ngrams(tokens, n)
                for ngram in ngrams:
                    if ngram not in seen_terms:
                        doc_freq[ngram] += 1
                        seen_terms.add(ngram)

        min_count = max(1, int(self.min_df))
        max_count = int(self.max_df * n_docs)

        vocab_idx = 0
        for term, count in sorted(doc_freq.items()):
            if min_count <= count <= max_count and vocab_idx < self.word_max:
                self.vocab[term] = vocab_idx
```

```
                self.idf[term] = math.log(n_docs / count)
                vocab_idx += 1

        return self

    def transform(self, texts):
        n_docs = len(texts)
        n_features = len(self.vocab)
        matrix = lil_matrix((n_docs, n_features))

        for doc_idx, text in enumerate(texts):
            tokens = self._tokenize(text)
            term_freq = defaultdict(int)

            for n in range(self.ngram_range[0], self.ngram_range[1] + 1):
                ngrams = self._get_ngrams(tokens, n)
                for ngram in ngrams:
                    if ngram in self.vocab:
                        term_freq[ngram] += 1

            for term, count in term_freq.items():
                term_idx = self.vocab[term]
                tf = 1 + math.log(count)
                tfidf_value = tf * self.idf[term]
                matrix[doc_idx, term_idx] = tfidf_value

        return matrix.tocsr()
```

[20]:
```
train_texts_prep = df.iloc[x_train]['tfidf_text'].values
test_texts_prep = df.iloc[x_test]['tfidf_text'].values
train_texts_raw = df.iloc[x_train]['raw_text'].values
test_texts_raw = df.iloc[x_test]['raw_text'].values

extractor_prep = TFIDFExtractor(word_max=35000)
extractor_prep.fit(train_texts_prep)
X_train_tfidf_prep = extractor_prep.transform(train_texts_prep)
X_test_tfidf_prep = extractor_prep.transform(test_texts_prep)

extractor_raw = TFIDFExtractor(word_max=35000)
extractor_raw.fit(train_texts_raw)
X_train_tfidf_raw = extractor_raw.transform(train_texts_raw)
X_test_tfidf_raw = extractor_raw.transform(test_texts_raw)

print(f"Preprocessed TF-IDF: {X_train_tfidf_prep.shape} (train),␣
↪{X_test_tfidf_prep.shape} (test)")
print(f"Raw TF-IDF:          {X_train_tfidf_raw.shape} (train),␣
↪{X_test_tfidf_raw.shape} (test)")
```

```
Preprocessed TF-IDF: (1400, 5394) (train), (600, 5394) (test)
Raw TF-IDF:          (1400, 7892) (train), (600, 7892) (test)
```

[21]:
```
results_prep = {}

print("TF-IDF (Preprocessed)\n")

for clf_name, clf in [('SVM', SVC(kernel='rbf')),
                      ('Logistic Regression',␣
 ↪LogisticRegression(max_iter=1000)),
                      ('KNN', KNeighborsClassifier(n_neighbors=5))]:
    clf.fit(X_train_tfidf_prep, y_train)
    y_pred = clf.predict(X_test_tfidf_prep)

    accuracy = accuracy_score(y_test, y_pred)

    results_prep[clf_name] = {'accuracy': accuracy}

    print(f"{clf_name:<25} Accuracy: {accuracy:.4f}")
```

```
TF-IDF (Preprocessed)

SVM                       Accuracy: 0.8967
Logistic Regression       Accuracy: 0.9117
KNN                       Accuracy: 0.4617
```

[22]:
```
y_test_array = y_test if isinstance(y_test, np.ndarray) else y_test.values
clf_tfidf = SVC(kernel='rbf')
clf_tfidf.fit(X_train_tfidf_prep, y_train)
y_pred_tfidf = clf_tfidf.predict(X_test_tfidf_prep)
analyze_misclassified("TF-IDF (Preprocessed)", y_pred_tfidf, x_test, y_test)
```

```
TF-IDF (Preprocessed)
--------------------------------------------------------------------------------
--------------------

Class 'account_access': 15 errors (showing 3)
1. Actual=account_access | Pred=bug_report | Text: Third-party OpenID provider
changed causing auth failure…
2. Actual=account_access | Pred=bug_report | Text: Inactivity timeout is set way
too short…
3. Actual=account_access | Pred=bug_report | Text: It worked yesterday. Today it
doesn't. Nothing has changed on my end…

Class 'billing': 15 errors (showing 3)
1. Actual=billing | Pred=bug_report | Text: Our finance department needs to
allocate costs to different departments but your system doesn't support cost
allocation…
```

```
2. Actual=billing | Pred=bug_report | Text: I need detailed hour-by-hour usage
data to reconcile charges but you only provide monthly summaries…
3. Actual=billing | Pred=bug_report | Text: My free trial ended and you
immediately charged without sending a reminder or confirmation…

Class 'bug_report': 11 errors (showing 3)
1. Actual=bug_report | Pred=account_access | Text: My two-factor codes keep
getting rejected as invalid. I enter them the moment they appear in the app so
they cannot have expired. I have been locked o…
2. Actual=bug_report | Pred=shipping_delivery | Text: I first reported this
issue three weeks ago and nothing has been done. The problem is still there
every day. I need a real fix, not a workaround, and …
3. Actual=bug_report | Pred=shipping_delivery | Text: When I try to place an
order theres no way to choose a delivery method. The dropdown is empty. I cant
complete my purchase. My address is in the deliv…

Class 'refund_request': 14 errors (showing 3)
1. Actual=refund_request | Pred=shipping_delivery | Text: My package was stolen
from my porch. Kindly reimbursement or reship to a safer location…
2. Actual=refund_request | Pred=bug_report | Text: I paid for somthing that
didnt work as expected and now im stuck trying to get it resolved. I've
contacted support beofre with no luck. Just need some…
3. Actual=refund_request | Pred=bug_report | Text: Something went wrong with my
recent order and I havent been able to get it sorted. Ive reached out before
without resolution. I just need someone to a…

Class 'shipping_delivery': 7 errors (showing 3)
1. Actual=shipping_delivery | Pred=refund_request | Text: You deducted the
return shipping cost from the credit on my account. I was not told this would
happen. The deduction made the credit much smaller than…
2. Actual=shipping_delivery | Pred=billing | Text: My international order got
hit with a customs fee that was never mentioned at checkout. This added
unexpected cost to my order and I wasnt prepared fo…
3. Actual=shipping_delivery | Pred=billing | Text: There is a line item on my
bill labeled handling fee that was never mentioned when I signed up. I did not
agree to this. Remove it and return the amou…
```

0.0.6 5. Word2Vec

[23]:
```python
def scale_dense_features(X_train, X_test):
    scaler = StandardScaler()
    return scaler.fit_transform(X_train), scaler.transform(X_test)

def get_avg_embedding(text, keyed_vectors, vector_size):
    words = text.split()
    vectors = [keyed_vectors[word] for word in words if word in keyed_vectors]
    if not vectors:
```

```
        return np.zeros(vector_size, dtype=np.float32)
    return np.mean(vectors, axis=0)

def build_embedding_feature_pair(df, train_idx, test_idx, keyed_vectors,␣
 ↪vector_size, prep_col='embedding_text', raw_col='raw_text'):
    X_train_prep = np.array([get_avg_embedding(df.iloc[i][prep_col],␣
 ↪keyed_vectors, vector_size) for i in train_idx])
    X_test_prep = np.array([get_avg_embedding(df.iloc[i][prep_col],␣
 ↪keyed_vectors, vector_size) for i in test_idx])
    X_train_raw = np.array([get_avg_embedding(df.iloc[i][raw_col],␣
 ↪keyed_vectors, vector_size) for i in train_idx])
    X_test_raw = np.array([get_avg_embedding(df.iloc[i][raw_col],␣
 ↪keyed_vectors, vector_size) for i in test_idx])

    X_train_prep, X_test_prep = scale_dense_features(X_train_prep, X_test_prep)
    X_train_raw, X_test_raw = scale_dense_features(X_train_raw, X_test_raw)
    return X_train_prep, X_test_prep, X_train_raw, X_test_raw

def build_tfidf_weighted_embeddings(texts, tfidf_extractor, keyed_vectors,␣
 ↪vector_size):
    X = np.zeros((len(texts), vector_size), dtype=np.float32)

    for doc_idx, text in enumerate(texts):
        tokens = re.findall(r'\b\w+\b', text.lower())
        term_freq = defaultdict(int)

        for n in range(tfidf_extractor.ngram_range[0], tfidf_extractor.
 ↪ngram_range[1] + 1):
            ngrams = [' '.join(tokens[i:i+n]) for i in range(len(tokens) - n +␣
 ↪1)]
            for ngram in ngrams:
                if ngram in tfidf_extractor.vocab:
                    term_freq[ngram] += 1

        weighted_vector = np.zeros(vector_size, dtype=np.float32)
        weight_sum = 0.0

        for term, count in term_freq.items():
            tf = 1 + math.log(count)
            idf = tfidf_extractor.idf[term]
            tfidf_score = tf * idf

            if term in keyed_vectors:
                weighted_vector += keyed_vectors[term] * tfidf_score
                weight_sum += tfidf_score
```

```
        if weight_sum > 0:
            weighted_vector /= weight_sum

        X[doc_idx] = weighted_vector

    X_scaled, _ = scale_dense_features(X, X)
    return X_scaled

def train_local_word2vec(sentences, vector_size=300, epochs=45):
    local_model = Word2Vec(
        sentences=sentences,
        vector_size=vector_size,
        window=5,
        min_count=1,
        sg=1,
        workers=4,
        epochs=epochs,
    )
    return local_model.wv

print("Loading pretrained Word2Vec (Google News 300d)...")
train_sentences_embedding = [df.iloc[i]['embedding_text'].split() for i in␣
 ↪x_train]
try:
    w2v_vectors = api.load("word2vec-google-news-300")
    w2v_dim = w2v_vectors.vector_size
    print(f"Pretrained Word2Vec loaded: {w2v_dim}-dimensional")
except Exception as ex:
    print(f"Pretrained Word2Vec unavailable ({ex}). Training local fallback␣
 ↪model...")
    w2v_vectors = train_local_word2vec(train_sentences_embedding,␣
 ↪vector_size=300, epochs=55)
    w2v_dim = w2v_vectors.vector_size
    print(f"Local Word2Vec trained: {w2v_dim}-dimensional")
```

```
Loading pretrained Word2Vec (Google News 300d)…
Pretrained Word2Vec loaded: 300-dimensional
```

[24]:
```
print("Creating Word2Vec vectors for preprocessed and raw text...")
X_train_embed_prep, X_test_embed_prep, X_train_embed_raw, X_test_embed_raw =␣
 ↪build_embedding_feature_pair(
    df,
    x_train,
    x_test,
    w2v_vectors,
    w2v_dim,
    prep_col='embedding_text',
```

```
    raw_col='raw_text',
)

print(f"Word2Vec (Preprocessed): {X_train_embed_prep.shape} (train),␣
↪{X_test_embed_prep.shape} (test)")
print(f"Word2Vec (Raw):         {X_train_embed_raw.shape} (train),␣
↪{X_test_embed_raw.shape} (test)")
```

```
Creating Word2Vec vectors for preprocessed and raw text…
Word2Vec (Preprocessed): (1400, 300) (train), (600, 300) (test)
Word2Vec (Raw):         (1400, 300) (train), (600, 300) (test)
```

[25]:
```
results_embed_prep = {}

print("Word2Vec (Preprocessed)\n")

for clf_name, clf in [('SVM', SVC(kernel='rbf')),
                      ('Logistic Regression',␣
↪LogisticRegression(max_iter=1000)),
                      ('KNN', KNeighborsClassifier(n_neighbors=5))]:
    clf.fit(X_train_embed_prep, y_train)
    y_pred = clf.predict(X_test_embed_prep)

    accuracy = accuracy_score(y_test, y_pred)

    results_embed_prep[clf_name] = {'accuracy': accuracy}

    print(f"{clf_name:<25} Accuracy: {accuracy:.4f}")
```

```
Word2Vec (Preprocessed)

SVM                       Accuracy: 0.8433
Logistic Regression       Accuracy: 0.8750
KNN                       Accuracy: 0.8650
```

[26]:
```
def analyze_misclassified(method_name, y_pred, X_test_indices, y_test_true,␣
↪max_per_class=3):
    import numpy as np

    y_test_array = y_test_true if isinstance(y_test_true, np.ndarray) else␣
↪y_test_true.values
    misclassified_idx = np.where(y_pred != y_test_array)[0]

    print(f"\n{method_name}")
    print("-" * 100)

    if len(misclassified_idx) == 0:
        print("No misclassified examples!")
```

```
        return

    for class_label in sorted(unique_labels):
        class_errors = [idx for idx in misclassified_idx if y_test_array[idx]␣
↪== class_label]
        if not class_errors: continue

        selected = class_errors[:max_per_class]
        print(f"\n{class_label}: {len(class_errors)} errors")

        for i, idx in enumerate(selected, 1):
            text = df.iloc[X_test_indices[idx]]['text'][:150].replace("\n", " ")
            print(f" {i}. Actual={y_test_array[idx]} | Pred={y_pred[idx]} |␣
↪Text: {text}")
```

[27]:
```
clf_w2v = SVC(kernel='rbf')
clf_w2v.fit(X_train_embed_prep, y_train)
y_pred_w2v = clf_w2v.predict(X_test_embed_prep)
analyze_misclassified("Word2Vec (Preprocessed)", y_pred_w2v, x_test, y_test)
```

```
Word2Vec (Preprocessed)
--------------------------------------------------------------------------------
--------------------

account_access: 25 errors
 1. Actual=account_access | Pred=bug_report | Text: Third-party OpenID provider
changed causing auth failure.
 2. Actual=account_access | Pred=bug_report | Text: Security questions are easy
to guess.
 3. Actual=account_access | Pred=billing | Text: My card was declined and
immediately my account was locked. I have since updated my payment details and
the new card works fine, but the account is st

billing: 16 errors
 1. Actual=billing | Pred=bug_report | Text: Our finance department needs to
allocate costs to different departments but your system doesn't support cost
allocation.
 2. Actual=billing | Pred=bug_report | Text: My free trial ended and you
immediately charged without sending a reminder or confirmation.
 3. Actual=billing | Pred=bug_report | Text: If competitors offer better pricing
will you match it?

bug_report: 15 errors
 1. Actual=bug_report | Pred=account_access | Text: When I search my own records
I am sometimes seeing entries that appear to belong to a different user. This is
a serious concern. I should only be able
 2. Actual=bug_report | Pred=billing | Text: When I go to billing settings to
```

```
add a new card the form just reloads blank after I submit. My card is never
saved. I cannot update my payment details
 3. Actual=bug_report | Pred=account_access | Text: I setup notifcations for new
activity but nothing is coming through. Checked spam, checked settings,
everything looks right. Feels like the account is

refund_request: 15 errors
 1. Actual=refund_request | Pred=billing | Text: I see a charge of $9.99 labeled
'Add-on Feature' on my bill but I never purchased any add-ons. Tell me what this
is for and return of my payment it.
 2. Actual=refund_request | Pred=shipping_delivery | Text: My package was stolen
from my porch. Kindly reimbursement or reship to a safer location.
 3. Actual=refund_request | Pred=billing | Text: I requested one reimbursement
but I see three separate reimbursement transactions from you in my bank account.
I'm confused about what I actually owe

shipping_delivery: 23 errors
 1. Actual=shipping_delivery | Pred=refund_request | Text: You deducted the
return shipping cost from the credit on my account. I was not told this would
happen. The deduction made the credit much smaller than
 2. Actual=shipping_delivery | Pred=refund_request | Text: I placed an order a
few days ago, received a payment confirmation, but I cant find the order
anywhere in my account. Nothing in order history. Its lik
 3. Actual=shipping_delivery | Pred=billing | Text: My international order got
hit with a customs fee that was never mentioned at checkout. This added
unexpected cost to my order and I wasnt prepared fo
```

0.0.7 6. Word2Vec + TF-IDF

[28]:
```
print("Word2Vec + TF-IDF (Preprocessed)\n")

X_train_w2v_tfidf_prep = build_tfidf_weighted_embeddings(train_texts_prep,␣
↪extractor_prep, w2v_vectors, w2v_dim)
X_test_w2v_tfidf_prep = build_tfidf_weighted_embeddings(test_texts_prep,␣
↪extractor_prep, w2v_vectors, w2v_dim)

results_w2v_tfidf_prep = {}

for clf_name, clf in [('SVM', SVC(kernel='rbf')),
                      ('Logistic Regression',␣
↪LogisticRegression(max_iter=1000)),
                      ('KNN', KNeighborsClassifier(n_neighbors=5))]:
    clf.fit(X_train_w2v_tfidf_prep, y_train)
    y_pred = clf.predict(X_test_w2v_tfidf_prep)

    accuracy = accuracy_score(y_test, y_pred)
    results_w2v_tfidf_prep[clf_name] = {'accuracy': accuracy}
```

```
    print(f"{clf_name:<25} Accuracy: {accuracy:.4f}")
```

```
Word2Vec + TF-IDF (Preprocessed)

SVM                       Accuracy: 0.8683
Logistic Regression       Accuracy: 0.8650
KNN                       Accuracy: 0.8417
```

[29]:
```
clf_w2v_tfidf = SVC(kernel='rbf')
clf_w2v_tfidf.fit(X_train_w2v_tfidf_prep, y_train)
y_pred_w2v_tfidf = clf_w2v_tfidf.predict(X_test_w2v_tfidf_prep)
analyze_misclassified("Word2Vec + TF-IDF", y_pred_w2v_tfidf, x_test, y_test)
```

```
Word2Vec + TF-IDF
--------------------------------------------------------------------------------
--------------------

account_access: 13 errors
 1. Actual=account_access | Pred=bug_report | Text: Security questions are easy
to guess.
 2. Actual=account_access | Pred=bug_report | Text: Inactivity timeout is set
way too short.
 3. Actual=account_access | Pred=refund_request | Text: It worked yesterday.
Today it doesn't. Nothing has changed on my end.

billing: 19 errors
 1. Actual=billing | Pred=bug_report | Text: I need detailed hour-by-hour usage
data to reconcile charges but you only provide monthly summaries.
 2. Actual=billing | Pred=bug_report | Text: My free trial ended and you
immediately charged without sending a reminder or confirmation.
 3. Actual=billing | Pred=bug_report | Text: You removed my payment method from
file without notifying me why.

bug_report: 20 errors
 1. Actual=bug_report | Pred=account_access | Text: I selected ten items for
deletion and confirmed. A completely different set of items was deleted. I lost
work I needed and the items I wanted to delet
 2. Actual=bug_report | Pred=billing | Text: When I go to billing settings to
add a new card the form just reloads blank after I submit. My card is never
saved. I cannot update my payment details
 3. Actual=bug_report | Pred=account_access | Text: I setup notifcations for new
activity but nothing is coming through. Checked spam, checked settings,
everything looks right. Feels like the account is

refund_request: 12 errors
 1. Actual=refund_request | Pred=billing | Text: I see a charge of $9.99 labeled
```

```
'Add-on Feature' on my bill but I never purchased any add-ons. Tell me what this
is for and return of my payment it.
 2. Actual=refund_request | Pred=shipping_delivery | Text: My package was stolen
from my porch. Kindly reimbursement or reship to a safer location.
 3. Actual=refund_request | Pred=account_access | Text: I paid for somthing that
didnt work as expected and now im stuck trying to get it resolved. I've
contacted support beofre with no luck. Just need some

shipping_delivery: 15 errors
 1. Actual=shipping_delivery | Pred=refund_request | Text: You deducted the
return shipping cost from the credit on my account. I was not told this would
happen. The deduction made the credit much smaller than
 2. Actual=shipping_delivery | Pred=refund_request | Text: I placed an order a
few days ago, received a payment confirmation, but I cant find the order
anywhere in my account. Nothing in order history. Its lik
 3. Actual=shipping_delivery | Pred=billing | Text: My international order got
hit with a customs fee that was never mentioned at checkout. This added
unexpected cost to my order and I wasnt prepared fo
```

0.0.8 7. Glove

[30]:
```python
print("Loading pretrained GloVe (glove-wiki-gigaword-100)...")

train_sentences_glove = [df.iloc[i]['embedding_text'].split() for i in x_train]

try:
    glove_vectors = api.load("glove-wiki-gigaword-100")
    glove_dim = glove_vectors.vector_size
    print(f"Pretrained GloVe loaded: {glove_dim}-dimensional")
except Exception as ex:
    print(f"Pretrained GloVe unavailable ({ex}). Training local fallback␣
 ↪vectors...")
    glove_vectors = train_local_word2vec(train_sentences_glove,␣
 ↪vector_size=100, epochs=45)
    glove_dim = glove_vectors.vector_size
    print(f"Local fallback vectors trained: {glove_dim}-dimensional")

print("Creating GloVe vectors for preprocessed text...")
X_train_glove_prep, X_test_glove_prep, _, _ = build_embedding_feature_pair(
    df,
    x_train,
    x_test,
    glove_vectors,
    glove_dim,
    prep_col='embedding_text',
)
```

```
print(f"GloVe (Preprocessed): {X_train_glove_prep.shape} (train),␣
↪{X_test_glove_prep.shape} (test)")

results_glove_prep = {}

print("\nGloVe (Preprocessed)\n")
for clf_name, clf in [('SVM', SVC(kernel='rbf')),
                      ('Logistic Regression',␣
↪LogisticRegression(max_iter=1000)),
                      ('KNN', KNeighborsClassifier(n_neighbors=5))]:
    clf.fit(X_train_glove_prep, y_train)
    y_pred = clf.predict(X_test_glove_prep)

    accuracy = accuracy_score(y_test, y_pred)
    results_glove_prep[clf_name] = {'accuracy': accuracy}

    print(f"{clf_name:<25} Accuracy: {accuracy:.4f}")
```

```
Loading pretrained GloVe (glove-wiki-gigaword-100)…
Pretrained GloVe loaded: 100-dimensional
Creating GloVe vectors for preprocessed text…
GloVe (Preprocessed): (1400, 100) (train), (600, 100) (test)

GloVe (Preprocessed)

SVM                       Accuracy: 0.8150
Logistic Regression       Accuracy: 0.8033
KNN                       Accuracy: 0.8133
```

[31]:
```
results_glove_prep = {}

print("GloVe (Preprocessed)\n")

for clf_name, clf in [('SVM', SVC(kernel='rbf')),
                      ('Logistic Regression',␣
↪LogisticRegression(max_iter=1000)),
                      ('KNN', KNeighborsClassifier(n_neighbors=5))]:
    clf.fit(X_train_glove_prep, y_train)
    y_pred = clf.predict(X_test_glove_prep)

    accuracy = accuracy_score(y_test, y_pred)
    results_glove_prep[clf_name] = {'accuracy': accuracy}

    print(f"{clf_name:<25} Accuracy: {accuracy:.4f}")
```

```
GloVe (Preprocessed)

SVM                       Accuracy: 0.8150
```

```
Logistic Regression       Accuracy: 0.8033
KNN                       Accuracy: 0.8133
```

[32]:
```python
clf_glove = SVC(kernel='rbf')
clf_glove.fit(X_train_glove_prep, y_train)
y_pred_glove = clf_glove.predict(X_test_glove_prep)
analyze_misclassified("GloVe (Preprocessed)", y_pred_glove, x_test, y_test)
```

```
GloVe (Preprocessed)
--------------------------------------------------------------------------------
--------------------

account_access: 24 errors
 1. Actual=account_access | Pred=bug_report | Text: Third-party OpenID provider
changed causing auth failure.
 2. Actual=account_access | Pred=bug_report | Text: Security questions are easy
to guess.
 3. Actual=account_access | Pred=billing | Text: My card was declined and
immediately my account was locked. I have since updated my payment details and
the new card works fine, but the account is st

billing: 21 errors
 1. Actual=billing | Pred=refund_request | Text: The volume commitment required
for discount doesn't match my actual needs.
 2. Actual=billing | Pred=bug_report | Text: Our finance department needs to
allocate costs to different departments but your system doesn't support cost
allocation.
 3. Actual=billing | Pred=bug_report | Text: I need detailed hour-by-hour usage
data to reconcile charges but you only provide monthly summaries.

bug_report: 22 errors
 1. Actual=bug_report | Pred=account_access | Text: When I search my own records
I am sometimes seeing entries that appear to belong to a different user. This is
a serious concern. I should only be able
 2. Actual=bug_report | Pred=billing | Text: When I go to billing settings to
add a new card the form just reloads blank after I submit. My card is never
saved. I cannot update my payment details
 3. Actual=bug_report | Pred=account_access | Text: The chat box opens but
messages don't send. The send button doesn't work no matter how many times I
click it.

refund_request: 20 errors
 1. Actual=refund_request | Pred=billing | Text: I see a charge of $9.99 labeled
'Add-on Feature' on my bill but I never purchased any add-ons. Tell me what this
is for and return of my payment it.
 2. Actual=refund_request | Pred=billing | Text: Rather than a cash payout I
would prefer you add the amount to my account as credit for future purchases.
```

```
Either way the overcharge needs to be correc
 3. Actual=refund_request | Pred=billing | Text: The product doesn't meet the
false advertising claims. I ask that you return what I paid.

shipping_delivery: 24 errors
 1. Actual=shipping_delivery | Pred=refund_request | Text: You deducted the
return shipping cost from the credit on my account. I was not told this would
happen. The deduction made the credit much smaller than
 2. Actual=shipping_delivery | Pred=refund_request | Text: I was told to return
my item but no return shipping label was included. Am I supposed to pay for
return shipping?
 3. Actual=shipping_delivery | Pred=bug_report | Text: My package appears to
have been opened and then carelessly repacked with tape everywhere. What
happened to it?
```

0.0.9 8. Glove + TF-IDF

[33]:
```
print("GloVe + TF-IDF (Preprocessed)\n")

X_train_glove_tfidf_prep = build_tfidf_weighted_embeddings(train_texts_prep,␣
 ↪extractor_prep, glove_vectors, glove_dim)
X_test_glove_tfidf_prep = build_tfidf_weighted_embeddings(test_texts_prep,␣
 ↪extractor_prep, glove_vectors, glove_dim)

results_glove_tfidf_prep = {}

for clf_name, clf in [('SVM', SVC(kernel='rbf')),
                      ('Logistic Regression',␣
 ↪LogisticRegression(max_iter=1000)),
                      ('KNN', KNeighborsClassifier(n_neighbors=5))]:
    clf.fit(X_train_glove_tfidf_prep, y_train)
    y_pred = clf.predict(X_test_glove_tfidf_prep)

    accuracy = accuracy_score(y_test, y_pred)
    results_glove_tfidf_prep[clf_name] = {'accuracy': accuracy}

    print(f"{clf_name:<25} Accuracy: {accuracy:.4f}")
```

```
GloVe + TF-IDF (Preprocessed)

SVM                       Accuracy: 0.8383
Logistic Regression       Accuracy: 0.8067
KNN                       Accuracy: 0.8000
```

[34]:
```
clf_glove_tfidf = SVC(kernel='rbf')
clf_glove_tfidf.fit(X_train_glove_tfidf_prep, y_train)
y_pred_glove_tfidf = clf_glove_tfidf.predict(X_test_glove_tfidf_prep)
analyze_misclassified("GloVe + TF-IDF", y_pred_glove_tfidf, x_test, y_test)
```

```
GloVe + TF-IDF
--------------------------------------------------------------------------------
--------------------

account_access: 18 errors
 1. Actual=account_access | Pred=bug_report | Text: Security questions are easy
to guess.
 2. Actual=account_access | Pred=billing | Text: My card was declined and
immediately my account was locked. I have since updated my payment details and
the new card works fine, but the account is st
 3. Actual=account_access | Pred=bug_report | Text: Inactivity timeout is set
way too short.

billing: 21 errors
 1. Actual=billing | Pred=refund_request | Text: Our finance department needs to
allocate costs to different departments but your system doesn't support cost
allocation.
 2. Actual=billing | Pred=refund_request | Text: My free trial ended and you
immediately charged without sending a reminder or confirmation.
 3. Actual=billing | Pred=account_access | Text: You removed my payment method
from file without notifying me why.

bug_report: 21 errors
 1. Actual=bug_report | Pred=account_access | Text: CSS rules aren't applying
because of specificity issues. The current behavior breaks the normal flow. I
need this fixed in the next release.
 2. Actual=bug_report | Pred=billing | Text: When I go to billing settings to
add a new card the form just reloads blank after I submit. My card is never
saved. I cannot update my payment details
 3. Actual=bug_report | Pred=shipping_delivery | Text: I selected 10 items to
delete and confirmed, but a completely different set of items was deleted.

refund_request: 18 errors
 1. Actual=refund_request | Pred=billing | Text: I'm posting about this on
social media to warn others. I only expected one payment for this order or
subscription. The issue shows up every time I rep
 2. Actual=refund_request | Pred=shipping_delivery | Text: My package was stolen
from my porch. Kindly reimbursement or reship to a safer location.
 3. Actual=refund_request | Pred=account_access | Text: I paid for somthing that
didnt work as expected and now im stuck trying to get it resolved. I've
contacted support beofre with no luck. Just need some

shipping_delivery: 19 errors
 1. Actual=shipping_delivery | Pred=refund_request | Text: You deducted the
return shipping cost from the credit on my account. I was not told this would
happen. The deduction made the credit much smaller than
 2. Actual=shipping_delivery | Pred=refund_request | Text: I received an email
```

```
saying my order was cancelled due to failed delivery but I never received any
delivery attempt. I need a replacement or redelivery
 3. Actual=shipping_delivery | Pred=billing | Text: I placed an order a few days
ago, received a payment confirmation, but I cant find the order anywhere in my
account. Nothing in order history. Its lik
```

0.0.10 9. FastText

[35]:
```
print("Training FastText embeddings (subword-aware)...")

train_sentences_fasttext_prep = [df.iloc[i]['fasttext_text'].split() for i in␣
 ↪x_train]
train_sentences_fasttext_raw = [df.iloc[i]['raw_text'].lower().split() for i in␣
 ↪x_train]

fasttext_prep = FastText(
    vector_size=220,
    window=5,
    min_count=1,
    sg=1,
    min_n=3,
    max_n=6,
    workers=4,
)
fasttext_prep.build_vocab(train_sentences_fasttext_prep)
fasttext_prep.train(train_sentences_fasttext_prep,␣
 ↪total_examples=len(train_sentences_fasttext_prep), epochs=55)

fasttext_raw = FastText(
    vector_size=220,
    window=5,
    min_count=1,
    sg=1,
    min_n=3,
    max_n=6,
    workers=4,
)
fasttext_raw.build_vocab(train_sentences_fasttext_raw)
fasttext_raw.train(train_sentences_fasttext_raw,␣
 ↪total_examples=len(train_sentences_fasttext_raw), epochs=55)

print("Creating FastText vectors for preprocessed and raw text...")
X_train_embed_ft_prep, X_test_embed_ft_prep, X_train_embed_ft_raw,␣
 ↪X_test_embed_ft_raw = build_embedding_feature_pair(
    df,
    x_train,
    x_test,
```

```
    fasttext_prep.wv,
    fasttext_prep.vector_size,
    prep_col='fasttext_text',
    raw_col='raw_text',
)

print(f"FastText (Preprocessed): {X_train_embed_ft_prep.shape} (train),␣
↪{X_test_embed_ft_prep.shape} (test)")
print(f"FastText (Raw):         {X_train_embed_ft_raw.shape} (train),␣
↪{X_test_embed_ft_raw.shape} (test)")
```

```
Training FastText embeddings (subword-aware)…

Exception ignored in: 'gensim.models.word2vec_inner.our_dot_float'

Creating FastText vectors for preprocessed and raw text…
FastText (Preprocessed): (1400, 220) (train), (600, 220) (test)
FastText (Raw):         (1400, 220) (train), (600, 220) (test)
```

[36]:
```
results_ft_prep = {}

print("FastText (Preprocessed)\n")

for clf_name, clf in [('SVM', SVC(kernel='rbf')),
                      ('Logistic Regression',␣
↪LogisticRegression(max_iter=1000)),
                      ('KNN', KNeighborsClassifier(n_neighbors=5))]:
    clf.fit(X_train_embed_ft_prep, y_train)
    y_pred = clf.predict(X_test_embed_ft_prep)

    accuracy = accuracy_score(y_test, y_pred)

    results_ft_prep[clf_name] = {'accuracy': accuracy}

    print(f"{clf_name:<25} Accuracy: {accuracy:.4f}")
```

```
FastText (Preprocessed)

SVM                       Accuracy: 0.9083
Logistic Regression       Accuracy: 0.8750
KNN                       Accuracy: 0.8833
```

[37]:
```
clf_ft = SVC(kernel='rbf')
clf_ft.fit(X_train_embed_ft_prep, y_train)
y_pred_ft = clf_ft.predict(X_test_embed_ft_prep)
analyze_misclassified("FastText (Preprocessed)", y_pred_ft, x_test, y_test)
```

```
FastText (Preprocessed)
--------------------------------------------------------------------------------
```

```
--------------------

account_access: 12 errors
 1. Actual=account_access | Pred=shipping_delivery | Text: Third-party OpenID
provider changed causing auth failure.
 2. Actual=account_access | Pred=billing | Text: My dashboard says my account is
suspended for non-payment yet my bank clearly shows the charge was processed
last Tuesday. Someone needs to reconcile
 3. Actual=account_access | Pred=bug_report | Text: I accidentally created two
accounts and now I have duplicates I can't merge.

billing: 8 errors
 1. Actual=billing | Pred=refund_request | Text: I want to be absolutely clear
that I am not asking for a refund or any money to be returned to me. I cannot
access my account dashboard to see the bre
 2. Actual=billing | Pred=refund_request | Text: My free trial ended and you
immediately charged without sending a reminder or confirmation.
 3. Actual=billing | Pred=refund_request | Text: I was charged and I want it
refunded. The service wasnt what was described and I dont want to continue.
Please process the refund to my original payme

bug_report: 18 errors
 1. Actual=bug_report | Pred=account_access | Text: I selected ten items for
deletion and confirmed. A completely different set of items was deleted. I lost
work I needed and the items I wanted to delet
 2. Actual=bug_report | Pred=shipping_delivery | Text: Array was mutated when it
shouldn't have been.
 3. Actual=bug_report | Pred=refund_request | Text: The checkout failed with an
error message but money came out of my account anyway. No order confirmation, no
product, but the charge is there. Please

refund_request: 11 errors
 1. Actual=refund_request | Pred=billing | Text: I see a charge of $9.99 labeled
'Add-on Feature' on my bill but I never purchased any add-ons. Tell me what this
is for and return of my payment it.
 2. Actual=refund_request | Pred=shipping_delivery | Text: My package was stolen
from my porch. Kindly reimbursement or reship to a safer location.
 3. Actual=refund_request | Pred=billing | Text: I upgraded to the annual plan
three days ago but it's not the right fit. I'd like to downgrade back to monthly
and get a prorated credit back for the

shipping_delivery: 6 errors
 1. Actual=shipping_delivery | Pred=refund_request | Text: You deducted the
return shipping cost from the credit on my account. I was not told this would
happen. The deduction made the credit much smaller than
 2. Actual=shipping_delivery | Pred=refund_request | Text: I placed an order a
few days ago, received a payment confirmation, but I cant find the order
anywhere in my account. Nothing in order history. Its lik
```

```
 3. Actual=shipping_delivery | Pred=billing | Text: There is a line item on my
bill labeled handling fee that was never mentioned when I signed up. I did not
agree to this. Remove it and return the amou
```

0.1 10. FastText + TF-IDF

[38]:
```
print("FastText + TF-IDF (Preprocessed)\n")

X_train_ft_tfidf_prep = build_tfidf_weighted_embeddings(train_texts_prep,␣
 ↪extractor_prep, fasttext_prep.wv, fasttext_prep.vector_size)
X_test_ft_tfidf_prep = build_tfidf_weighted_embeddings(test_texts_prep,␣
 ↪extractor_prep, fasttext_prep.wv, fasttext_prep.vector_size)

results_ft_tfidf_prep = {}

for clf_name, clf in [('SVM', SVC(kernel='rbf')),
                      ('Logistic Regression',␣
 ↪LogisticRegression(max_iter=1000)),
                      ('KNN', KNeighborsClassifier(n_neighbors=5))]:
    clf.fit(X_train_ft_tfidf_prep, y_train)
    y_pred = clf.predict(X_test_ft_tfidf_prep)

    accuracy = accuracy_score(y_test, y_pred)
    results_ft_tfidf_prep[clf_name] = {'accuracy': accuracy}

    print(f"{clf_name:<25} Accuracy: {accuracy:.4f}")
```

```
FastText + TF-IDF (Preprocessed)

SVM                       Accuracy: 0.9100
Logistic Regression       Accuracy: 0.8817
KNN                       Accuracy: 0.8850
```

[39]:
```
clf_ft_tfidf = SVC(kernel='rbf')
clf_ft_tfidf.fit(X_train_ft_tfidf_prep, y_train)
y_pred_ft_tfidf = clf_ft_tfidf.predict(X_test_ft_tfidf_prep)
analyze_misclassified("FastText + TF-IDF", y_pred_ft_tfidf, x_test, y_test)
```

```
FastText + TF-IDF
--------------------------------------------------------------------------------
--------------------

account_access: 9 errors
 1. Actual=account_access | Pred=bug_report | Text: Session timeout happens at
random intarvals.
 2. Actual=account_access | Pred=shipping_delivery | Text: I can't figure out
how to appeal my account suspension.
```

```
 3. Actual=account_access | Pred=bug_report | Text: Multiple concurrent logins
cause session conflicts.

billing: 10 errors
 1. Actual=billing | Pred=refund_request | Text: My free trial ended and you
immediately charged without sending a reminder or confirmation.
 2. Actual=billing | Pred=refund_request | Text: I was charged and I want it
refunded. The service wasnt what was described and I dont want to continue.
Please process the refund to my original payme
 3. Actual=billing | Pred=bug_report | Text: If a payment is late you're
charging interest that seems to be compounding more than it should.

bug_report: 17 errors
 1. Actual=bug_report | Pred=account_access | Text: I selected ten items for
deletion and confirmed. A completely different set of items was deleted. I lost
work I needed and the items I wanted to delet
 2. Actual=bug_report | Pred=refund_request | Text: The checkout failed with an
error message but money came out of my account anyway. No order confirmation, no
product, but the charge is there. Please
 3. Actual=bug_report | Pred=refund_request | Text: I selected 10 items to
delete and confirmed, but a completely different set of items was deleted.

refund_request: 12 errors
 1. Actual=refund_request | Pred=billing | Text: I see a charge of $9.99 labeled
'Add-on Feature' on my bill but I never purchased any add-ons. Tell me what this
is for and return of my payment it.
 2. Actual=refund_request | Pred=billing | Text: I requested one reimbursement
but I see three separate reimbursement transactions from you in my bank account.
I'm confused about what I actually owe
 3. Actual=refund_request | Pred=billing | Text: I upgraded to the annual plan
three days ago but it's not the right fit. I'd like to downgrade back to monthly
and get a prorated credit back for the

shipping_delivery: 6 errors
 1. Actual=shipping_delivery | Pred=refund_request | Text: I placed an order a
few days ago, received a payment confirmation, but I cant find the order
anywhere in my account. Nothing in order history. Its lik
 2. Actual=shipping_delivery | Pred=billing | Text: My international order got
hit with a customs fee that was never mentioned at checkout. This added
unexpected cost to my order and I wasnt prepared fo
 3. Actual=shipping_delivery | Pred=billing | Text: There is a line item on my
bill labeled handling fee that was never mentioned when I signed up. I did not
agree to this. Remove it and return the amou
```

0.1.1 11. Bert Features + Classification

[40]:
```
tokenizer_bert = BertTokenizer.from_pretrained('bert-base-uncased')
bert_model = BertModel.from_pretrained('bert-base-uncased')
bert_model.to(device)
bert_model.eval()

bert_batch_size = 24 if device.type == 'cuda' else 10
bert_max_length = 192

print(f"BERT loaded (output dimension: 768, batch_size={bert_batch_size},␣
↪max_length={bert_max_length})")

def extract_bert_features(texts, tokenizer, model, device, batch_size=16,␣
↪max_length=192):
    features = []

    with torch.no_grad():
        for i in range(0, len(texts), batch_size):
            batch_texts = texts[i:i + batch_size]

            encodings = tokenizer(
                batch_texts,
                add_special_tokens=True,
                max_length=max_length,
                padding=True,
                truncation=True,
                return_tensors='pt'
            )

            input_ids = encodings['input_ids'].to(device)
            attention_mask = encodings['attention_mask'].to(device)

            outputs = model(input_ids, attention_mask=attention_mask)
            token_embeddings = outputs.last_hidden_state
            mask_expanded = attention_mask.unsqueeze(-1).float()
            sum_embeddings = (token_embeddings * mask_expanded).sum(dim=1)
            sum_mask = mask_expanded.sum(dim=1).clamp(min=1e-9)
            batch_features = (sum_embeddings / sum_mask).cpu().numpy()
            features.extend(batch_features)

    return np.array(features)
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a
HF_TOKEN to enable higher rate limits and faster downloads.

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: bert-base-uncased
```

```
Key                                          | Status     |  |
---------------------------------------------+------------+--+-
cls.predictions.bias                         | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.bias     | UNEXPECTED |  |
cls.predictions.transform.dense.bias         | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight   | UNEXPECTED |  |
cls.predictions.transform.dense.weight       | UNEXPECTED |  |
cls.seq_relationship.weight                  | UNEXPECTED |  |
cls.seq_relationship.bias                    | UNEXPECTED |  |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architecture;
not ok if you expect identical arch.
```

```
BERT loaded (output dimension: 768, batch_size=10, max_length=192)
```

[41]:
```
print("Extracting BERT features from preprocessed text...")
train_texts_bert_prep = df.iloc[x_train]['bert_text'].tolist()
test_texts_bert_prep = df.iloc[x_test]['bert_text'].tolist()
X_train_embed_bert_prep = extract_bert_features(
    train_texts_bert_prep, tokenizer_bert, bert_model, device,␣
 ↪batch_size=bert_batch_size, max_length=bert_max_length
)
X_test_embed_bert_prep = extract_bert_features(
    test_texts_bert_prep, tokenizer_bert, bert_model, device,␣
 ↪batch_size=bert_batch_size, max_length=bert_max_length
)
X_train_embed_bert_prep, X_test_embed_bert_prep =␣
 ↪scale_dense_features(X_train_embed_bert_prep, X_test_embed_bert_prep)

print("Extracting BERT features from raw text...")
train_texts_bert_raw = df.iloc[x_train]['raw_text'].tolist()
test_texts_bert_raw = df.iloc[x_test]['raw_text'].tolist()
X_train_embed_bert_raw = extract_bert_features(
    train_texts_bert_raw, tokenizer_bert, bert_model, device,␣
 ↪batch_size=bert_batch_size, max_length=bert_max_length
)
X_test_embed_bert_raw = extract_bert_features(
    test_texts_bert_raw, tokenizer_bert, bert_model, device,␣
 ↪batch_size=bert_batch_size, max_length=bert_max_length
)
X_train_embed_bert_raw, X_test_embed_bert_raw =␣
 ↪scale_dense_features(X_train_embed_bert_raw, X_test_embed_bert_raw)

print(f"BERT (Preprocessed): {X_train_embed_bert_prep.shape} (train),␣
 ↪{X_test_embed_bert_prep.shape} (test)")
```

```
print(f"BERT (Raw):         {X_train_embed_bert_raw.shape} (train),␣
↪{X_test_embed_bert_raw.shape} (test)")
```

```
Extracting BERT features from preprocessed text…
Extracting BERT features from raw text…
BERT (Preprocessed): (1400, 768) (train), (600, 768) (test)
BERT (Raw):         (1400, 768) (train), (600, 768) (test)
```

[42]:
```
results_bert_raw = {}

print("BERT (Raw)\n")

for clf_name, clf in [('SVM', SVC(kernel='rbf')),
                      ('Logistic Regression',␣
↪LogisticRegression(max_iter=1000)),
                      ('KNN', KNeighborsClassifier(n_neighbors=5))]:
    clf.fit(X_train_embed_bert_raw, y_train)
    y_pred = clf.predict(X_test_embed_bert_raw)

    accuracy = accuracy_score(y_test, y_pred)

    results_bert_raw[clf_name] = {'accuracy': accuracy}

    print(f"{clf_name:<25} Accuracy: {accuracy:.4f}")
```

```
BERT (Raw)

SVM                       Accuracy: 0.8717
Logistic Regression       Accuracy: 0.8650
KNN                       Accuracy: 0.8500
```

[43]:
```
clf_bert = SVC(kernel='rbf')
clf_bert.fit(X_train_embed_bert_raw, y_train)
y_pred_bert = clf_bert.predict(X_test_embed_bert_raw)
analyze_misclassified("BERT (Raw)", y_pred_bert, x_test, y_test)
```

```
BERT (Raw)
--------------------------------------------------------------------------------
--------------------

account_access: 14 errors
 1. Actual=account_access | Pred=bug_report | Text: Security questions are easy
to guess.
 2. Actual=account_access | Pred=bug_report | Text: Inactivity timeout is set
way too short.
 3. Actual=account_access | Pred=bug_report | Text: Face ID or fingerprint auth
works sometimes but not others.
```

```
billing: 13 errors
 1. Actual=billing | Pred=shipping_delivery | Text: The volume commitment
required for discount doesn't match my actual needs.
 2. Actual=billing | Pred=refund_request | Text: My free trial ended and you
immediately charged without sending a reminder or confirmation.
 3. Actual=billing | Pred=bug_report | Text: If competitors offer better pricing
will you match it?

bug_report: 21 errors
 1. Actual=bug_report | Pred=account_access | Text: I selected ten items for
deletion and confirmed. A completely different set of items was deleted. I lost
work I needed and the items I wanted to delet
 2. Actual=bug_report | Pred=account_access | Text: When I search my own records
I am sometimes seeing entries that appear to belong to a different user. This is
a serious concern. I should only be able
 3. Actual=bug_report | Pred=billing | Text: When I go to billing settings to
add a new card the form just reloads blank after I submit. My card is never
saved. I cannot update my payment details

refund_request: 16 errors
 1. Actual=refund_request | Pred=billing | Text: I paid for somthing that didnt
work as expected and now im stuck trying to get it resolved. I've contacted
support beofre with no luck. Just need some
 2. Actual=refund_request | Pred=shipping_delivery | Text: Something went wrong
with my recent order and I havent been able to get it sorted. Ive reached out
before without resolution. I just need someone to a
 3. Actual=refund_request | Pred=billing | Text: I requested one reimbursement
but I see three separate reimbursement transactions from you in my bank account.
I'm confused about what I actually owe

shipping_delivery: 13 errors
 1. Actual=shipping_delivery | Pred=refund_request | Text: You deducted the
return shipping cost from the credit on my account. I was not told this would
happen. The deduction made the credit much smaller than
 2. Actual=shipping_delivery | Pred=billing | Text: I placed an order a few days
ago, received a payment confirmation, but I cant find the order anywhere in my
account. Nothing in order history. Its lik
 3. Actual=shipping_delivery | Pred=billing | Text: I want to waive signature
requirement but can't.
```

0.1.2 12. Bert Finetuning

[44]:
```python
from torch.utils.data import Dataset, DataLoader
from transformers import BertTokenizerFast, BertForSequenceClassification
from torch.optim import AdamW
import torch
import numpy as np
```

```python
from sklearn.metrics import accuracy_score

# Dataset
class TicketDataset(Dataset):
    def __init__(self, texts, labels, tokenizer, max_length=192):
        self.encodings = tokenizer(
            texts,
            truncation=True,
            padding=True,
            max_length=max_length
        )
        self.labels = labels

    def __getitem__(self, idx):
        item = {k: torch.tensor(v[idx]) for k, v in self.encodings.items()}
        item['labels'] = torch.tensor(self.labels[idx])
        return item

    def __len__(self):
        return len(self.labels)

# Prepare data
train_texts = df.iloc[x_train]['bert_text'].tolist()
test_texts = df.iloc[x_test]['bert_text'].tolist()

label_to_id = {label: i for i, label in enumerate(unique_labels)}
id_to_label = {i: label for label, i in label_to_id.items()}

y_train_ids = np.array([label_to_id[y] for y in y_train])
y_test_ids = np.array([label_to_id[y] for y in y_test])

# Tokenizer + Model
tokenizer = BertTokenizerFast.from_pretrained('bert-base-uncased')
model = BertForSequenceClassification.from_pretrained(
    'bert-base-uncased',
    num_labels=len(unique_labels)
)

model.to(device)

# DataLoaders
train_dataset = TicketDataset(train_texts, y_train_ids, tokenizer)
test_dataset = TicketDataset(test_texts, y_test_ids, tokenizer)

train_loader = DataLoader(train_dataset, batch_size=12, shuffle=True)
test_loader = DataLoader(test_dataset, batch_size=12)
```

```python
# Optimizer
optimizer = AdamW(model.parameters(), lr=2e-5)

epochs = 3

for epoch in range(epochs):
    # ===== TRAIN =====
    model.train()
    train_loss = 0
    train_preds = []
    train_true = []

    for batch in train_loader:
        optimizer.zero_grad()

        input_ids = batch['input_ids'].to(device)
        attention_mask = batch['attention_mask'].to(device)
        labels = batch['labels'].to(device)

        outputs = model(
            input_ids=input_ids,
            attention_mask=attention_mask,
            labels=labels
        )

        loss = outputs.loss
        logits = outputs.logits

        loss.backward()
        optimizer.step()

        train_loss += loss.item()

        preds = torch.argmax(logits, dim=1).detach().cpu().numpy()
        train_preds.extend(preds)
        train_true.extend(labels.detach().cpu().numpy())

    train_loss /= len(train_loader)
    train_acc = accuracy_score(train_true, train_preds)

    # ===== EVAL =====
    model.eval()
    test_loss = 0
    test_preds = []
    test_true = []

    with torch.no_grad():
```

```
        for batch in test_loader:
            input_ids = batch['input_ids'].to(device)
            attention_mask = batch['attention_mask'].to(device)
            labels = batch['labels'].to(device)

            outputs = model(
                input_ids=input_ids,
                attention_mask=attention_mask,
                labels=labels
            )

            loss = outputs.loss
            logits = outputs.logits

            test_loss += loss.item()

            preds = torch.argmax(logits, dim=1).cpu().numpy()
            test_preds.extend(preds)
            test_true.extend(labels.cpu().numpy())

    test_loss /= len(test_loader)
    test_acc = accuracy_score(test_true, test_preds)

    print(f"Epoch {epoch+1}/{epochs} | Train Loss: {train_loss:.4f}")

# Final predictions for analysis
y_pred_labels = np.array([id_to_label[i] for i in test_preds])
```

```
Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]
```

```
BertForSequenceClassification LOAD REPORT from: bert-base-uncased
Key                                          | Status     |
---------------------------------------------+------------+-
cls.predictions.bias                         | UNEXPECTED |
cls.predictions.transform.LayerNorm.bias     | UNEXPECTED |
cls.predictions.transform.dense.bias         | UNEXPECTED |
cls.predictions.transform.LayerNorm.weight   | UNEXPECTED |
cls.predictions.transform.dense.weight       | UNEXPECTED |
cls.seq_relationship.weight                  | UNEXPECTED |
cls.seq_relationship.bias                    | UNEXPECTED |
classifier.weight                            | MISSING    |
classifier.bias                              | MISSING    |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architecture;
not ok if you expect identical arch.
- MISSING:      those params were newly initialized because missing from the
checkpoint. Consider training on your downstream task.
```

```
Epoch 1/3 | Train Loss: 0.8825
Epoch 2/3 | Train Loss: 0.2898
Epoch 3/3 | Train Loss: 0.1474
```

[51]:
```python
model.eval()

train_preds = []
train_true = []

with torch.no_grad():
    for batch in train_loader:
        input_ids = batch['input_ids'].to(device)
        attention_mask = batch['attention_mask'].to(device)
        labels = batch['labels'].to(device)

        outputs = model(input_ids=input_ids, attention_mask=attention_mask)
        logits = outputs.logits

        preds = torch.argmax(logits, dim=1).cpu().numpy()
        train_preds.extend(preds)
        train_true.extend(labels.cpu().numpy())

train_acc = accuracy_score(train_true, train_preds)


test_preds = []
test_true = []

with torch.no_grad():
    for batch in test_loader:
        input_ids = batch['input_ids'].to(device)
        attention_mask = batch['attention_mask'].to(device)
        labels = batch['labels'].to(device)

        outputs = model(input_ids=input_ids, attention_mask=attention_mask)
        logits = outputs.logits

        preds = torch.argmax(logits, dim=1).cpu().numpy()
        test_preds.extend(preds)
        test_true.extend(labels.cpu().numpy())

test_acc = accuracy_score(test_true, test_preds)

print(f"Test  Accuracy: {test_acc:.4f}")

results_bert_ft = {
    'BERT': {'accuracy': test_acc}
```

```
}
```

```
Test  Accuracy: 0.9333
```

[52]:
```
analyze_misclassified(
    "BERT (Fine-tuned)",
    y_pred_labels,
    x_test,
    y_test
)
```

```
BERT (Fine-tuned)
--------------------------------------------------------------------------------
--------------------

account_access: 8 errors
 1. Actual=account_access | Pred=billing | Text: My card was declined and
immediately my account was locked. I have since updated my payment details and
the new card works fine, but the account is st
 2. Actual=account_access | Pred=refund_request | Text: It worked yesterday.
Today it doesn't. Nothing has changed on my end.
 3. Actual=account_access | Pred=billing | Text: My dashboard says my account is
suspended for non-payment yet my bank clearly shows the charge was processed
last Tuesday. Someone needs to reconcile

billing: 9 errors
 1. Actual=billing | Pred=refund_request | Text: My free trial ended and you
immediately charged without sending a reminder or confirmation.
 2. Actual=billing | Pred=bug_report | Text: The line items add up to one total
but the statement shows a different total. The math doesn't work out.
 3. Actual=billing | Pred=bug_report | Text: My bank shows I was charged for Pro
plan but the app says I'm on Free. Where did my money go?

bug_report: 20 errors
 1. Actual=bug_report | Pred=account_access | Text: I selected ten items for
deletion and confirmed. A completely different set of items was deleted. I lost
work I needed and the items I wanted to delet
 2. Actual=bug_report | Pred=account_access | Text: When I search my own records
I am sometimes seeing entries that appear to belong to a different user. This is
a serious concern. I should only be able
 3. Actual=bug_report | Pred=account_access | Text: When syncing between devices
sometimes data gets corrupted if there are conflicts.

refund_request: 12 errors
 1. Actual=refund_request | Pred=billing | Text: I see a charge of $9.99 labeled
'Add-on Feature' on my bill but I never purchased any add-ons. Tell me what this
is for and return of my payment it.
```

```
 2. Actual=refund_request | Pred=billing | Text: I paid for somthing that didnt
work as expected and now im stuck trying to get it resolved. I've contacted
support beofre with no luck. Just need some
 3. Actual=refund_request | Pred=shipping_delivery | Text: Something went wrong
with my recent order and I havent been able to get it sorted. Ive reached out
before without resolution. I just need someone to a

shipping_delivery: 9 errors
 1. Actual=shipping_delivery | Pred=refund_request | Text: You deducted the
return shipping cost from the credit on my account. I was not told this would
happen. The deduction made the credit much smaller than
 2. Actual=shipping_delivery | Pred=refund_request | Text: I was told to return
my item but no return shipping label was included. Am I supposed to pay for
return shipping?
 3. Actual=shipping_delivery | Pred=account_access | Text: I want to waive
signature requirement but can't.
```

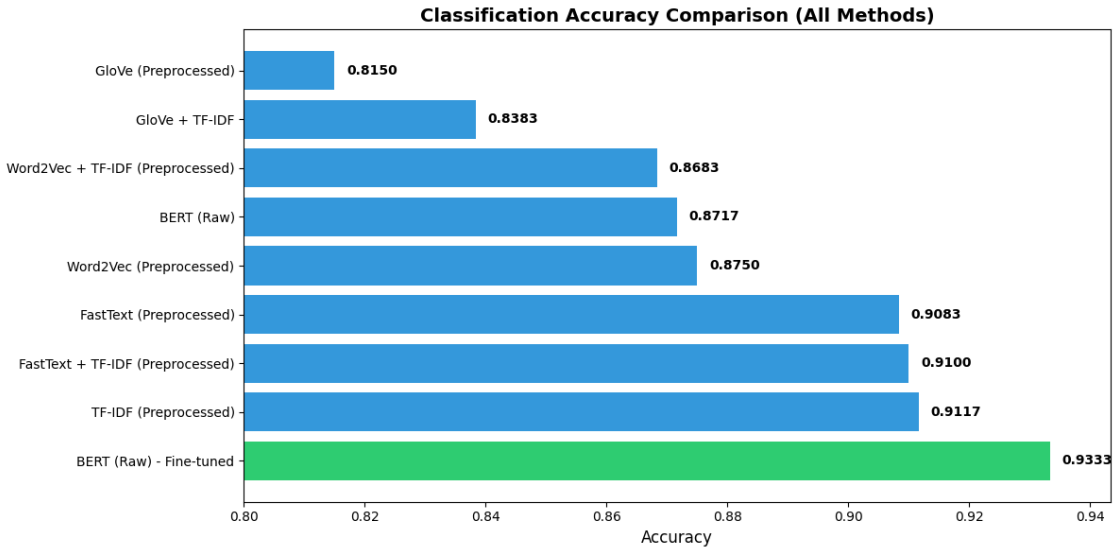
[53]:
```python
import pandas as pd

all_results = {
    'TF-IDF (Preprocessed)': results_prep,
    'Word2Vec (Preprocessed)': results_embed_prep,
    'Word2Vec + TF-IDF (Preprocessed)': results_w2v_tfidf_prep,
    'GloVe (Preprocessed)': results_glove_prep,
    'GloVe + TF-IDF': results_glove_tfidf_prep,
    'FastText (Preprocessed)': results_ft_prep,
    'FastText + TF-IDF (Preprocessed)': results_ft_tfidf_prep,
    'BERT (Raw)': results_bert_raw,
    'BERT (Raw) - Fine-tuned': results_bert_ft,
}

best_results = {}

print("="*80)
print("BEST CLASSIFIER FOR EACH METHOD")
print("="*80)

for method_name, results in all_results.items():
    best_clf = max(results.items(), key=lambda x: x[1]['accuracy'])
    best_results[method_name] = {'classifier': best_clf[0], 'accuracy':␣
 ↪best_clf[1]['accuracy']}
    print(f"{method_name:35s} → {best_clf[0]:20s} (Acc:␣
 ↪{best_clf[1]['accuracy']:.4f})")

print("\n" + "="*80)
print("OVERALL BEST: {:.4f} ({})".format(
    max(best_results.values(), key=lambda x: x['accuracy'])['accuracy'],
```

```
    max(best_results.items(), key=lambda x: x[1]['accuracy'])[0]
))
print("="*80)
```

```
================================================================================
BEST CLASSIFIER FOR EACH METHOD
================================================================================
TF-IDF (Preprocessed)               → Logistic Regression  (Acc: 0.9117)
Word2Vec (Preprocessed)             → Logistic Regression  (Acc: 0.8750)
Word2Vec + TF-IDF (Preprocessed)    → SVM                  (Acc: 0.8683)
GloVe (Preprocessed)                → SVM                  (Acc: 0.8150)
GloVe + TF-IDF                      → SVM                  (Acc: 0.8383)
FastText (Preprocessed)             → SVM                  (Acc: 0.9083)
FastText + TF-IDF (Preprocessed)    → SVM                  (Acc: 0.9100)
BERT (Raw)                          → SVM                  (Acc: 0.8717)
BERT (Raw) - Fine-tuned             → BERT                 (Acc: 0.9333)

================================================================================
OVERALL BEST: 0.9333 (BERT (Raw) - Fine-tuned)
================================================================================
```

[54]:
```
import matplotlib.pyplot as plt
import seaborn as sns

results_df = pd.DataFrame([
    {'Method': method, 'Accuracy': data['accuracy']}
    for method, data in best_results.items()
]).sort_values('Accuracy', ascending=False)

fig, ax = plt.subplots(figsize=(12, 6))
colors = ['#2ecc71' if x == results_df['Accuracy'].max() else '#3498db' for x␣
 ↪in results_df['Accuracy']]
bars = ax.barh(results_df['Method'], results_df['Accuracy'], color=colors)
ax.set_xlabel('Accuracy', fontsize=12)
ax.set_title('Classification Accuracy Comparison (All Methods)', fontsize=14,␣
 ↪fontweight='bold')
ax.set_xlim([0.8, max(results_df['Accuracy']) + 0.01])

for i, bar in enumerate(bars):
    width = bar.get_width()
    ax.text(width + 0.002, bar.get_y() + bar.get_height()/2,
            f'{width:.4f}', ha='left', va='center', fontsize=10,␣
 ↪fontweight='bold')

plt.tight_layout()
plt.show()
```

```
print("\n" + "="*80)
print("FINAL ACCURACY RESULTS (Sorted by Performance)")
print("="*80)
for idx, row in results_df.iterrows():
    print(f"{row['Method']:35s} → {row['Accuracy']:.4f}")
print("="*80)
```




```
================================================================================
FINAL ACCURACY RESULTS (Sorted by Performance)
================================================================================
BERT (Raw) - Fine-tuned             → 0.9333
TF-IDF (Preprocessed)               → 0.9117
FastText + TF-IDF (Preprocessed)    → 0.9100
FastText (Preprocessed)             → 0.9083
Word2Vec (Preprocessed)             → 0.8750
BERT (Raw)                          → 0.8717
Word2Vec + TF-IDF (Preprocessed)    → 0.8683
GloVe + TF-IDF                      → 0.8383
GloVe (Preprocessed)                → 0.8150
================================================================================
```

[55]:
```
print("\n" + "="*100)
print("COMPLETE ACCURACY RESULTS SUMMARY")
print("="*100)

summary_data = [
    (method, data['accuracy'], data['classifier'])
    for method, data in best_results.items()
```

```
]

print(f"\n{'Method':<35} {'Accuracy':<12} {'Best Classifier':<20}")
print("-"*100)

sorted_results = sorted(summary_data, key=lambda x: x[1], reverse=True)
for method, accuracy, classifier in sorted_results:
    marker = "   BEST" if accuracy == max(x[1] for x in sorted_results) else ""
    print(f"{method:<35} {accuracy:<12.4f} {classifier:<20}{marker}")

print("\n" + "="*100)
best_method = max(best_results.items(), key=lambda x: x[1]['accuracy'])
print(f"BEST OVERALL RESULT: {best_method[0]} achieves␣
 ↪{best_method[1]['accuracy']:.4f} accuracy")
print("="*100)
```

```
====================================================================================
====================
COMPLETE ACCURACY RESULTS SUMMARY
====================================================================================
====================

Method                                Accuracy     Best Classifier
------------------------------------------------------------------------------------
--------------------
BERT (Raw) - Fine-tuned               0.9333       BERT                    BEST
TF-IDF (Preprocessed)                 0.9117       Logistic Regression
FastText + TF-IDF (Preprocessed)      0.9100       SVM
FastText (Preprocessed)               0.9083       SVM
Word2Vec (Preprocessed)               0.8750       Logistic Regression
BERT (Raw)                            0.8717       SVM
Word2Vec + TF-IDF (Preprocessed)      0.8683       SVM
GloVe + TF-IDF                        0.8383       SVM
GloVe (Preprocessed)                  0.8150       SVM

====================================================================================
====================
BEST OVERALL RESULT: BERT (Raw) - Fine-tuned achieves 0.9333 accuracy
====================================================================================
====================
```

[60]:
```
def predict_ticket(text, model, tokenizer, id_to_label, device, max_length=192):
    model.eval()

    inputs = tokenizer(
        text,
```

```
        return_tensors='pt',
        truncation=True,
        padding=True,
        max_length=max_length
    )

    input_ids = inputs['input_ids'].to(device)
    attention_mask = inputs['attention_mask'].to(device)

    with torch.no_grad():
        outputs = model(input_ids=input_ids, attention_mask=attention_mask)
        logits = outputs.logits

    probs = torch.softmax(logits, dim=1).cpu().numpy()[0]
    pred_id = np.argmax(probs)

    return {
        "label": id_to_label[pred_id],
        "confidence": float(probs[pred_id]),
        "all_probs": {
            id_to_label[i]: float(probs[i]) for i in range(len(probs))
        }
    }

sample = "billing not working"

predict_ticket(sample, model, tokenizer, id_to_label, device)
```

[60]:
```
{'label': 'billing',
 'confidence': 0.824088990688324,
 'all_probs': {'account_access': 0.04302271082997322,
  'billing': 0.824088990688324,
  'bug_report': 0.0105444910004735,
  'refund_request': 0.09365485608577728,
  'shipping_delivery': 0.028688907623291016}}
```